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Accelerating Molecular Generation through Representation Alignment

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I, Shreyas Ravishankar of Hughes Hall, being a candidate for the Master of Philosophy in Advanced Computer Science, hereby declare that this report and the work described in it are my own work, unaided except as may be specified below, and that the report does not contain material that has already been used to any substantial extent for a comparable purpose. In preparation of this report, I adhered to the Department of Computer Science and Technology AI Policy. [The project required the approved use of {insert name of technology} and such use is acknowledged in the text at the relevant sections.] [I am content for my report to be made available to the students and staff of the University.]

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Abstract

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Chapter 1

Introduction

The evolution of generative machine learning in scientific domains is defined by the tension between data scale, data quality, and structural priors. Our report examines a recent addition to this story in *representation alignment*. We study this technique in two regimes for de novo molecular generation — small molecules and protein backbones — and characterise its behaviour. In this introduction, we trace the narrative arc of research trends in molecular generation, and highlight the motivation for our work: the emerging need for data-efficient modelling in a world of prior-light models.

The dominant ideas in generative modelling are *denoising diffusion* and its offshoot *flow-matching* [1, 2], which found breakthrough success in the realm of image generation. This paradigm has since been adapted for de novo design challenges across scientific fields, such as generating protein backbones [15, 16, 11], inorganic materials [18], and drug-like small molecules [12]. Training these models is typically expensive in both data and compute, with state-of-the-art image generators routinely training on billions of examples [24]. However, the corpus of available high-quality molecular data sits orders of magnitude below this. Experimental determination is slow and expensive, which means datasets are correspondingly small. For example, protein structure generators have largely been trained on at most half a million structures [11], which may explain why they have not yet matched the maturity of their vision-domain counterparts.

Historically, researchers compensated for the small scale of high-quality molecular data by encoding structural priors directly into model architectures [30]. SE(3)-equivariant architectures, for example, enforce the rotational and translational symmetries expected of molecules by construction [31]. But hard structural priors come with a computational cost, and the operations required to encode geometry are a bottleneck to scaling models.

A recent wave of research has moved entirely the other way: drop hard priors, use bare-bones non-equivariant transformer stacks, and rely on data scale and augmentation to teach the model what equivariance buys. This shift fits a broader trend in machine learning on leveraging representation instead of model architecture to inject soft inductive

biases for structured data. Vision Transformers [32] showed that the spatial-locality and translation-equivariance priors of convolutional neural nets (CNNs) can be learned rather than imposed, given sufficient scale. Graph Transformers like TokenGT [33] have made the same case for graphs. AlphaFold 3 [35] stripped many of the SE(3)-equivariant biases of AlphaFold 2 [34], replacing the structure module with a diffusion model operating on raw coordinates. Brehmer et al. [36] make the empirical scaling case directly: more data through a simpler model beats less data through a richer one. In our work, we examine Tabasco [12] and Proteina [11], which are flow-based generators built on vanilla transformers for small molecules and protein backbones respectively.

Given the difficulty of procuring data in scientific domains, data scale must necessarily come from non-experimental sources, which may be of lower quality. For example, recent efforts in protein generation have leaned on synthetic structures from folding-model predictions [11, 26] to expand training datasets into the tens of millions. However, even these enhanced corpora are arguably insufficient. Metagenomic databases catalogue over two billion natural protein sequences [28], and the theoretical sequence space at a median protein length [29] is hundreds of orders of magnitude greater still. Even at the frontier of the field’s data scale, we sample but a sliver of the distribution we wish to model.

In summary, the field has traded hard structural priors for architectural simplicity, and curated data for synthetic scale. However, this trade-off dilutes the structural signal available to a model. Furthermore, even our best attempts at creating large datasets may not be large enough. These twin challenges amplify the need for techniques that make training still more efficient without sacrificing quality.

But why is training diffusion-transformer architectures expensive in the first place? One answer comes from Yu et al. [3], who argue that these models suffer from a *representation bottleneck*. During training, a model is implicitly asked to learn two distinct tasks simultaneously: (1) representation: extracting semantically meaningful encodings from complex data; and (2) generation: constructing data from these encodings. The standard denoising objective alone may not provide sufficient signal for representation learning, leading to slow convergence and the large training budgets these models require in practice. In the context of molecular generation, this representation burden is compounded by the twin challenges above. Models are now asked to encode complex structural geometry, in a regime where high-quality data is scarce relative to true underlying distribution, without the aid of architectural priors.

Yu et al. propose Representation Alignment (REPA) as a remedy. The technique adds a regularisation term that aligns a generator’s intermediate hidden states with the embeddings of a frozen pretrained encoder. On image diffusion, they report roughly a $17.5\times$ training speed-up against a strong baseline. REPA has since been extended beyond images to video [7], audio [8], and recently to molecular generation [5, 6]. However, whether REPA’s gains transfer to molecular settings, and why, remain uncharacterised.

In this light, the core question we investigate in this report is: *can prior-light molecular generation models benefit from representation alignment with pre-trained models, in training efficiency and generation quality?* We aim to characterise when and why this works, including failure modes and limitations.

Our report documents two lines of enquiry.

- (i) *Does REPA work in this domain?*: We conduct empirical studies on training flow-based molecular generator architectures with REPA in two molecular generation regimes: Tabasco [12] for small molecules, and Proteina [11] for protein backbones. We find that REPA does not measurably improve Tabasco, but it improves training efficiency and advances the generation quality envelope for Proteina.
- (ii) *When does REPA work?*: Following Singh et al. [4] in the realm of image generation, we develop a profiling framework to characterise the suitability of an encoder as a representation target. We measure three properties of an encoder: (1) an upper bound on the structural information it contains (linear-probe recoverability against domain structural labels); (2) a lower bound on what a projector can transfer from that information (projector-based recoverability); (3) the geometric conditioning of its embeddings for the alignment objective (effective rank).

We do not propose a new architecture, alignment loss, or evaluation metric, and our two studies are not a controlled experiment over the conditions that govern REPA’s effectiveness. Instead, they are best read as paired case studies in regimes where those conditions differ.

The remainder of this report is organised as follows. Chapter 2 sets out the background needed to read the rest: flow matching, REPA, and the parameterisations we use for small molecules and proteins. Chapter 3 places the work in context, covering the scaling-over-priors trend in structural biology and the recent literature on representation alignment. Chapter 4 discusses what it means for a generative model to be “good” in our domain, and the tensions that question hides. Chapter 5 develops the encoder-profiling framework and uses it to predict where REPA may and may not help. Chapters 6 and 7 present the two paired studies, small molecules first and protein backbones second. Finally, we conclude in Chapter 8 with a discussion of whether what we learn generalises beyond molecular generation.

Chapter 2

Background

Chapter 3

Related work

Chapter 4

Evaluating molecular generation

Chapter 5

Encoder targets and profiling

Chapter 6

REPA on small molecules

Chapter 7

REPA on protein backbones

Chapter 8

Conclusions

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Appendix A

Technical details

A.1 Placeholder section

TODO.